

CHAPTER 8

Electrons in a Periodic Crystal

Contents

- ❑ Behaviours of electrons in crystalline solids
- ❑ Concepts of electronic band structure, effective mass, etc
- ❑ Electronic state of semiconductor quantum structures such as quantum wells and quantum dots
- ❑ Carrier localization due to disorder



8.0 One-electron approximation

$$H\phi(\mathbf{r}_i, \mathbf{R}_j) = E\phi(\mathbf{r}_i, \mathbf{R}_j)$$

$$H = -\frac{\hbar^2}{2} \sum_{j=1}^M \frac{1}{M_j} \Delta_{\mathbf{R}_j} - \frac{\hbar^2}{2m_0} \sum_{i=1}^N \Delta_{\mathbf{r}_i} + \frac{1}{4\pi\epsilon_0} \times \left(\sum_{j>j'} \frac{e^2 Z_j Z_{j'}}{|\mathbf{R}_j - \mathbf{R}_{j'}|} + \sum_{i>i'} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_{i'}|} + \sum_{i,j} \frac{e^2 Z_j}{|\mathbf{R}_j - \mathbf{r}_i|} \right)$$

□ Adiabatic approximation:

Separation of the motion of electrons from that of atoms

$$\phi(\mathbf{r}_i, \mathbf{R}_j) = \phi(\mathbf{r}_i) \phi(\mathbf{R}_{j0})$$

$$H(\mathbf{R}_j) \phi(\mathbf{R}_j) = E \phi(\mathbf{R}_j) \quad H_e(\mathbf{r}_i) \phi(\mathbf{r}_i) = E \phi(\mathbf{r}_i)$$

$$H = -\frac{\hbar^2}{2} \sum_{j=1}^M \frac{1}{M_j} \Delta_{\mathbf{R}_j} + \frac{1}{4\pi\epsilon_0} \sum_{j>j'} \frac{e^2 Z_j Z_{j'}}{|\mathbf{R}_j - \mathbf{R}_{j'}|}$$

$$H_e = -\frac{\hbar^2}{2m_0} \sum_{i=1}^N \Delta_{\mathbf{r}_i} + \frac{1}{4\pi\epsilon_0} \times \left(\sum_{i>i'} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_{i'}|} + \sum_{i,j} \frac{e^2 Z_j}{|\mathbf{R}_j - \mathbf{r}_i|} \right)$$



□ Single electron approximation:

$$H_{1,e}(\mathbf{r})\phi(\mathbf{r}) = E\phi(\mathbf{r})$$

$$\left[\frac{\hat{p}^2}{2m} + V(\mathbf{r})\right]\phi(\mathbf{r}) = E\phi(\mathbf{r})$$

$$\hat{p} = -i\hbar\nabla$$

$$V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$$

8.1 Bloch's Theorem and Bloch function

- When a particle moves in a periodic potential, its wavefunction can be expressed in a form known as Bloch function

$$\phi_{k,i}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{k,i}(\mathbf{r}), \quad (8.3a)$$

$$u_{k,i}(\mathbf{r}) = u_{k,i}(\mathbf{r} + \mathbf{R}). \quad (8.3b)$$

$\phi_{k,i}(\mathbf{r})$ s known as Bloch function. When $\phi_{k,i}$ is multiplied with $\exp(-i\omega t)$, it represents a plane wave whose amplitude is modulated by the periodic function $u_{k,i}(\mathbf{r})$

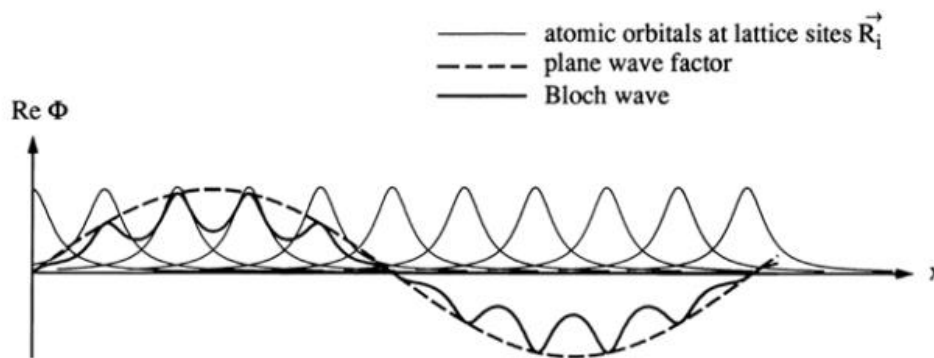


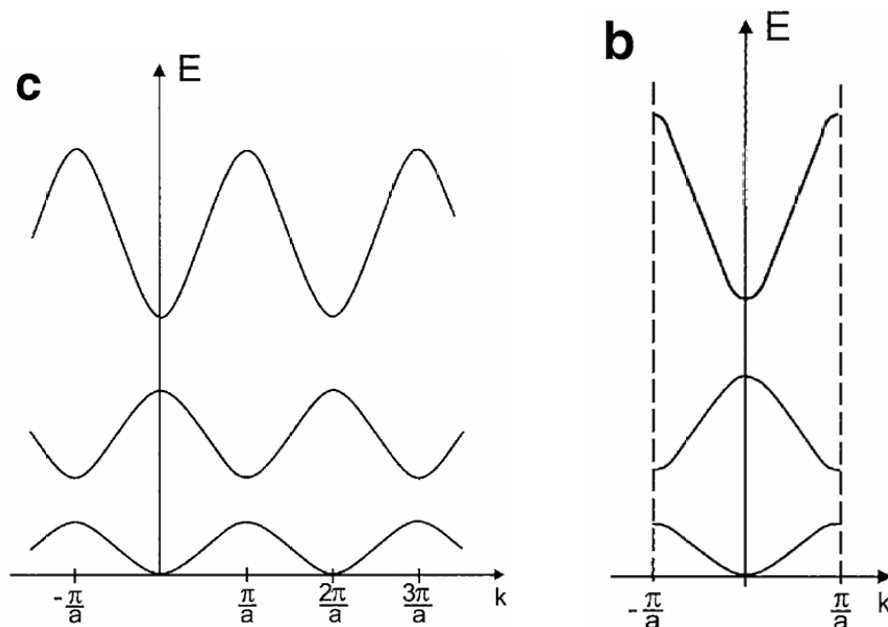
Fig. 8.2. Schematic drawing of the real (or imaginary) part of a Bloch wave in one dimension

□ The eigenenergies depend both on $\mathbf{k}$ and i and are periodic in $\mathbf{k}$ -space

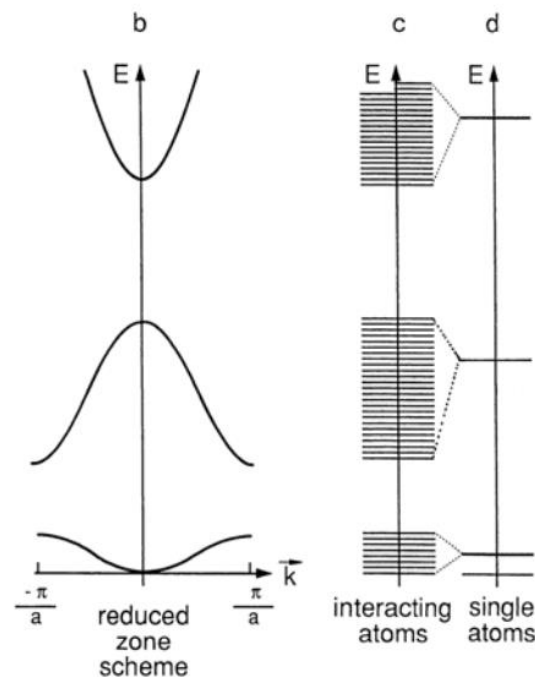
$$E_i(\mathbf{k}) = E_i(\mathbf{k} + \mathbf{G}) \quad (8.4)$$

$$\phi_{\mathbf{k},i}(\mathbf{r}) = \phi_{\mathbf{k}+\mathbf{G},i}(\mathbf{r}). \quad (8.5)$$

□ The eigenenergies form energy bands



The periodic (c) and the reduced (b) representation of the electronic band structure of a crystal



When atoms form solids, the energy levels of electrons in atoms are extended to energy bands

Band structure calculation

- Nearly free electron approach-free electron + weak periodic potential
- Linear combination of atomic orbitals (LCAO)
- Density functional theory (DFT) – First principles theory
 - ✓ Density functional theory
 - ✓ Local density functional theory
 - ✓ Kohn-Sham equation

Kohn is founder of density functional theory, and Pople is a pioneer in quantum chemistry and calculational chemistry. They shared the Nobel prize in chemistry in 1998.



Photo from the Nobel
Foundation archive.

Walter Kohn

Prize share: 1/2

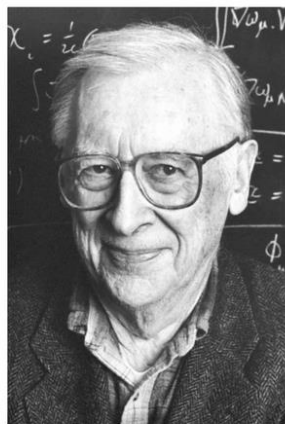


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John A. Pople

Prize share: 1/2

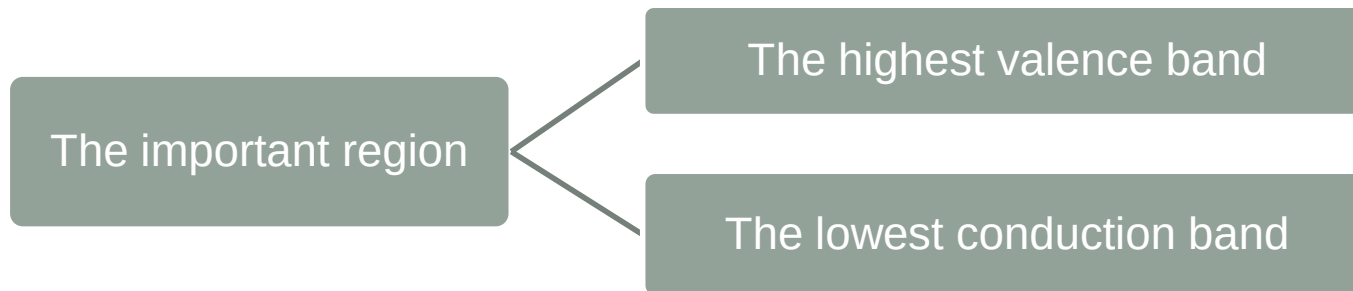
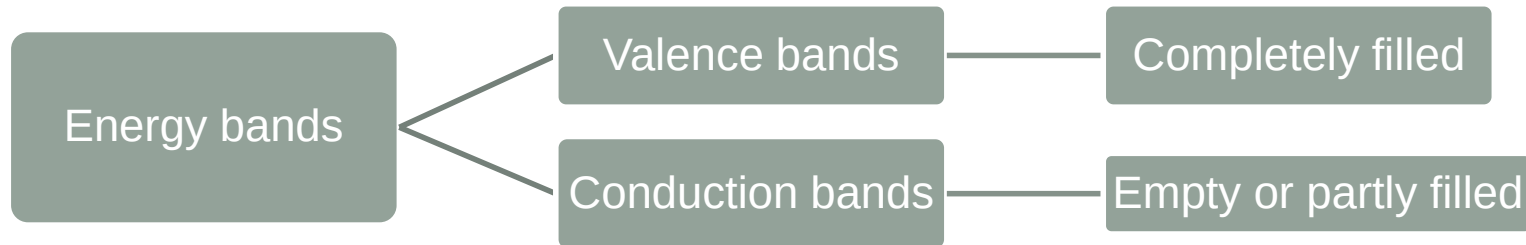


David Baker

大卫·贝克在哈佛大学学习的是哲学和社会科学专业，1989年获加利福尼亚大学伯克利分校博士学位。1993年，大卫·贝克就职于美国华盛顿大学西雅图分校，开始研究蛋白质结构和功能的计算方法，以及蛋白质设计，2024年获得诺贝尔化学奖。

8.2 Metals, Semiconductors, Insulators

Using Fermi-Dirac statistics, put all electrons into the bands. Assume $T=0\text{ K}$



□ The important region determines:

- The optical properties around the fundamental absorption edge
- The magnetic properties and the electronic contributions to the conductivities of electricity and of heat.

- E_g : The gap between CB & VB

Semiconductor : $0 < E_g \leq 4eV$

Insulator : $E_g \geq 4eV$

Semimetals : $E_g = 0eV$

- Semiconductors:

narrow gap: $0 < E_g \leq 0.5eV$

Usual gap: $0.5 < E_g \leq 2eV$

Wide gap: $2 < E_g \leq 4eV$

- Direct and indirect Semiconductors

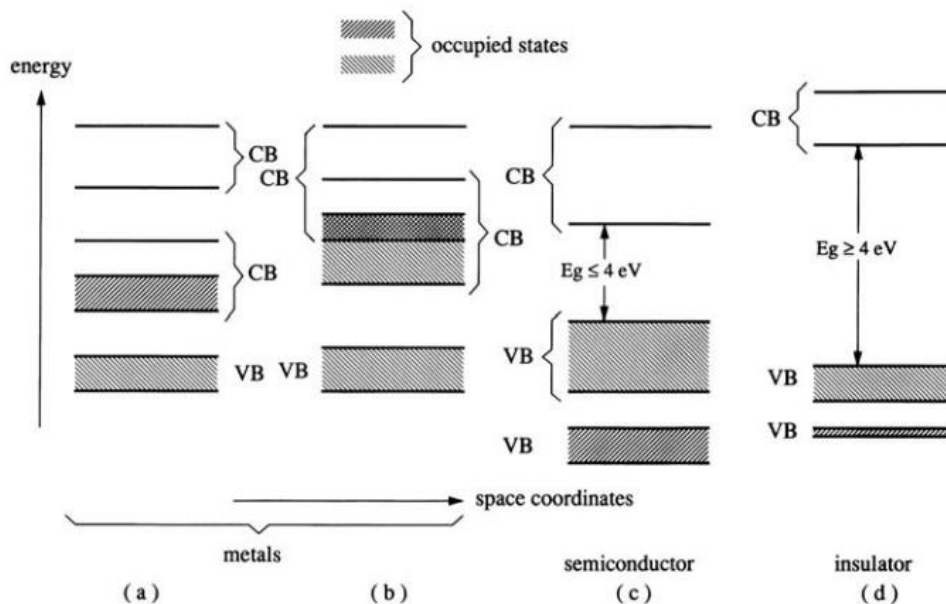
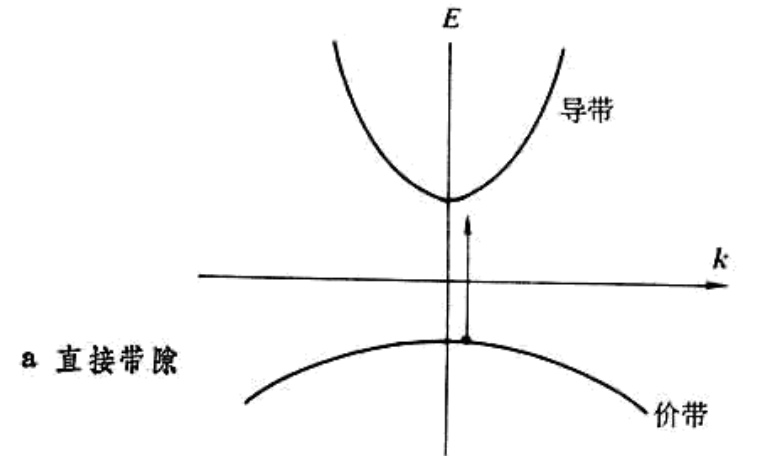
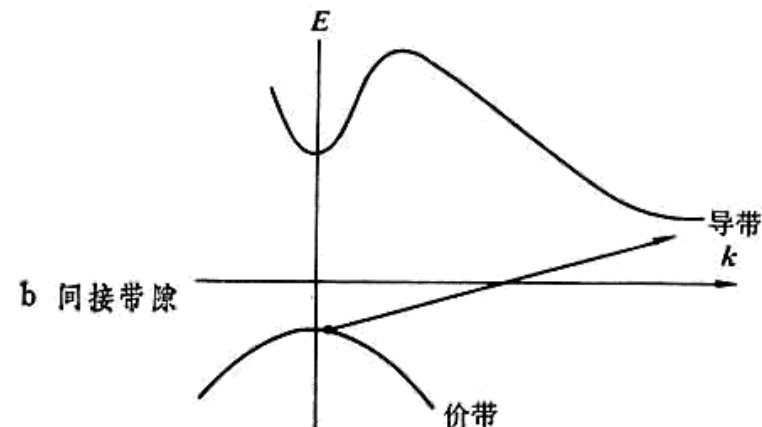


Fig. 8.5. Occupation of the bands for a metal (a,b) a semiconductor (c) and an insulator (d)



a 直接带隙



b 间接带隙

- Presently more than 600 elements and compound semiconductors are known.
- The most important is Si: $E_g \sim 1.1\text{eV}$
- SiC: IV semiconductors . Forming many different polytypes.
- C: graphite, fullerene (C_{60}), graphene carbon nanotube, graphdiyne
- Table.8.2 some group IV, III–V, II–VI and I–VII semiconductors
- Some ternary semiconductors: CuGaSe_2
- Quaternary ones like $\text{Ag}_2\text{CdGeS}_4$
- Alloys even without a miscibility gap:
 $\text{Si}_{1-x}\text{Ge}_x$, $\text{Ga}_{1-y}\text{Al}_y\text{As}$, $\text{CdS}_{1-x}\text{Se}_x$,
 $\text{ZnSe}_{1-x}\text{Te}_x$ or $\text{Cd}_{1-y}\text{Hg}_y\text{Te}$

I ^a	II ^a	III ^b	IV ^b	V ^b	VI ^b	VII ^b	VIII ^b			I ^b	II ^b	III ^a	IV ^a	V ^a	VI ^a	VII ^a	VIII ^a
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mb	43 Tc	44 Ru	45 Rh	46 PoL	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
																	
		58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu		
																	
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111	112	113	114	115	116	117	118
		90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr		

Table 8.2. Values of the band gap energy of some group IV, III-V, II-VI and I-VII semiconductors (SC) at low temperature ($T \lesssim 10$ K), Sy: Symmetry (point group). According to [82L1] of Chap. 1

Group IV				Group III-V				Group II-VI				Group I-VII			
SC	Sy	E_g (eV)	dir/indir	SC	Sy	E_g (eV)	dir/indir	SC	Sy	E_g (eV)	dir/indir	SC	Sy	E_g (eV)	dir/indir
C	O_h	5.48	i	AlN	C_{6v}	6.28	d	ZnO	C_{6v}	3.437	d	CuCl	T_d	3.395	d
Si	O_h	1.17	i	AlP	T_d	2.53	i	ZnS	C_{6v}	3.91	d	CuBr	T_d	3.077	d
Ge	O_h	0.744	i	AlAs	T_d	2.228	i	ZnS	T_d	3.78	d	CuI	T_d	3.115	d
grey Sn	O_h	0	semimetal	AlSb	T_d	1.696	i	ZnSe	T_d	2.82	d	AgCl	T_d	3.249	i
				GaN	C_{6v}	3.503	d	ZnTe	T_d	2.391	d	AgBr	T_d	2.684	i
				GaP	T_d	2.350	i	CdO	O_h	0.8	i	AgI	C_{6v}	3.024	d
				GaAs	T_d	1.518	d	CdS	C_{6v}	2.583	d				
				GaSb	T_d	0.812	d	CdSe	C_{6v}	1.841	d				
				InN	C_{6v}	2.2	d	CdTe	T_d	1.60	d				
				InP	T_d	1.424	d	HgS	T_d	0	d				
				InAs	T_d	0.418	d	HgSe	T_d	0	d				
				InSb	T_d	0.237	d	HgTe	T_d	0	d				

- A general trend: the band gap of semiconductors decreases with increasing temperature.

$$E_g(T) = E_g(T = 0) - \frac{\alpha T^2}{\beta + \gamma T}$$

$$\Delta E_g(T) = E_g(T = 0) - E_g(T) = \frac{\alpha T^2}{\beta + \gamma T} \quad (8.8)$$

8.4 Electrons and Holes in Crystals as New Quasiparticles

- Holes: “defect-electrons”, remove electrons from the valence band .
The electrons and holes exist only in the crystal.

Table 8.3. Properties of a hole in the valence band compared to the properties of the electron that has been removed from the valence band to create the hole

Property	Hole	Removed electron
electric charge q	q_h	$= -q_{re}, q_{re} \approx -1.6 \times 10^{-19} \text{ As}$
wave vector	$\mathbf{k}_h$	$= -\mathbf{k}_{re}$
spin	σ_h	$= -\sigma_{re}$
eff. mass	$m_h > 0$	$= -m_{re}, m_{re} < 0$

The energy of free electrons:

$$E_{e,p} = \pm \left(m_0 c^2 + \frac{\hbar^2 k^2}{2m_0} \right) \quad (8.10)$$

The electrons and holes are quasi particles in a semiconductor

The energy of a hole increases if it is brought deeper into the valence band.

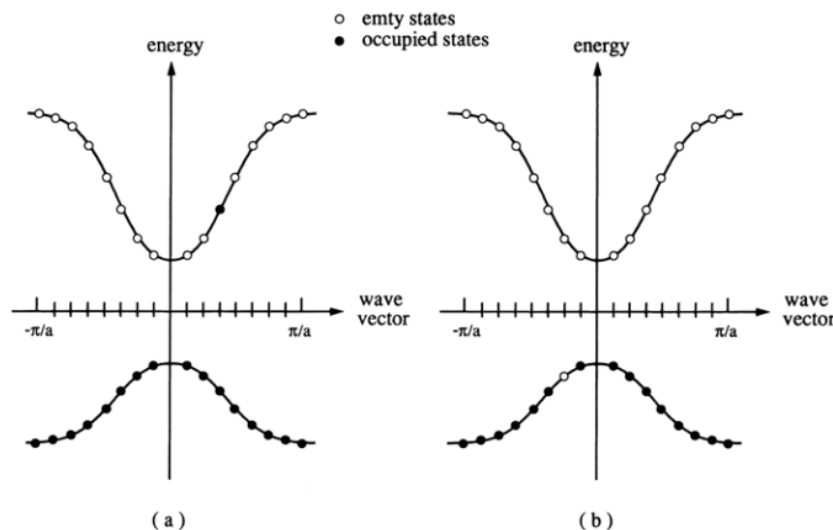


Fig. 8.6. One electron in the conduction band (a) and one hole in the valence band (b). Full circles: occupied states; open circles: empty states

8.5 The Effective-Mass Concept

□ The motion of an electron or hole in a semiconductor under an external field

An external force F changes the energy of the particle:

$$dE(\mathbf{k}) = \mathbf{F} \cdot d\mathbf{s} = \mathbf{F} \cdot \mathbf{v}_g dt \quad (8.12)$$

$$\mathbf{v}_g = \frac{1}{\hbar} \text{grad}_{\mathbf{k}} E(\mathbf{k}) \quad (8.13a)$$

or simplified:
$$v_g = \frac{1}{\hbar} \frac{dE}{dk} \quad (8.13b)$$

$$dE(\mathbf{k}) = \frac{dE(\mathbf{k})}{dk} \cdot d\mathbf{k} = \hbar v_g d\mathbf{k} \quad (8.13c)$$

Combining (8.12) and (8.13):
$$\hbar \frac{d\mathbf{k}}{dt} = \mathbf{F} = \dot{\mathbf{p}} \quad (8.13d)$$

So the acceleration a of the particle:

$$\mathbf{a} = \frac{d\mathbf{v}_g}{dt} = \frac{1}{\hbar} \frac{\partial^2 E}{\partial k \partial t} = \frac{1}{\hbar} \frac{\partial^2 E}{\partial k^2} \frac{d\mathbf{k}}{dt} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k^2} \mathbf{F} \quad (8.13e)$$

Considering
$$\mathbf{a} = \frac{1}{m} \mathbf{F} \quad (8.13f)$$

we get the effective mass :
$$\frac{1}{m_{eff}} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k^2} = \frac{1}{\hbar} \frac{\partial^2 E}{\partial k_i \partial k_j}; \quad i, j = x, y, z \quad (8.14)$$

- The effective mass is actually a tensor and can depend on the direction in which the electron or hole moves.
 - From Fig.8.7: The effective masses are constant in the vicinity of the band extrema.
 - Electron easy to move and accelerate $\longleftrightarrow$ low effective mass
 $\longleftrightarrow$ large curvature of the band $\longleftrightarrow$ large band width
 $\longleftrightarrow$ strong coupling between adjacent atoms and vice versa
- Narrow gap semiconductors $\longleftrightarrow$ low effective mass
Wide gap semiconductors $\longleftrightarrow$ large effective mass

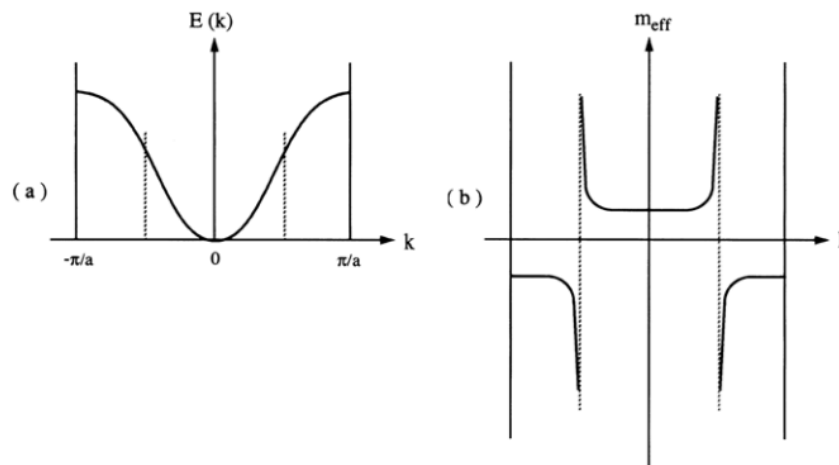


Fig. 8.7. Schematic dispersion (a) of a conduction band and the resulting effective mass (b). After [95I1] of Chap. 1

8.6 The Polaron Concept and Other Electron–Phonon Interaction Processes

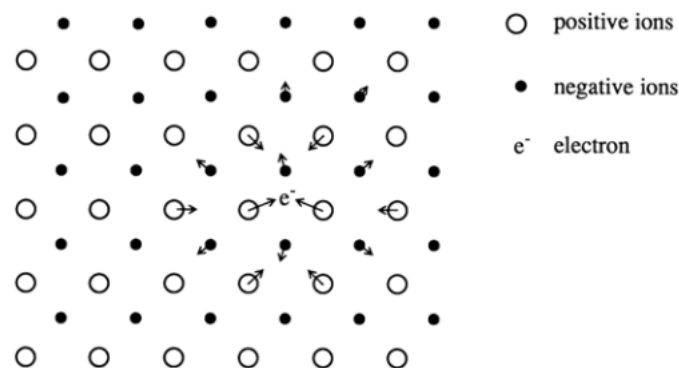
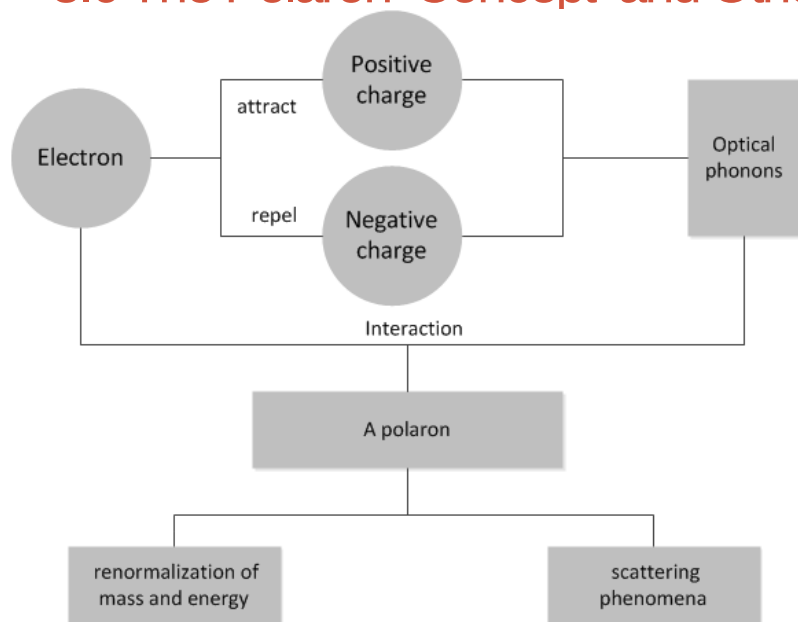


Fig. 8.8. The lattice distortion around a carrier in a (partly) ionic semiconductor illustrating the polaron concept

- The effective mass $m_{e,h}$ of a polaron is greater than that of an electron in a rigid lattice $m_{e,h}^{r.l.}$:

$$m_{e,h} \simeq m_{e,h}^{r.l.} \left(1 + \frac{\alpha_{e,h}}{6} \right) \quad (8.19a)$$

$$\alpha_{e,h} = \frac{e^2}{8\pi\epsilon_0\hbar\omega_{LO}} \left(\frac{2m_{e,h}^{r.l.}\omega_{LO}}{\hbar} \right) \left(\frac{1}{\epsilon_b} - \frac{1}{\epsilon_s} \right) \quad (8.19b)$$

Additionally the lattice relaxation leads to a decrease of the width of the gap by :

$$\Delta E_g^{e,h} = \alpha_{e,h} \hbar\omega_{LO} \quad (8.19d)$$

The radius of the phonon cloud in the polaron :

$$r_p^{e,h} = \left(\frac{\hbar}{2m_{e,h}^{r.l.}\omega_{LO}} \right)^{\frac{1}{2}} \quad (8.19e)$$

- An electron or hole can be scattered inelastically:

$$E_e^i = E_e^f \pm \hbar\omega_{phonon} \quad (8.20a)$$

$$\mathbf{k}_e^i = \mathbf{k}_e^f \pm \mathbf{k}_{phonon} (\pm \mathbf{G}) \quad (8.20b)$$

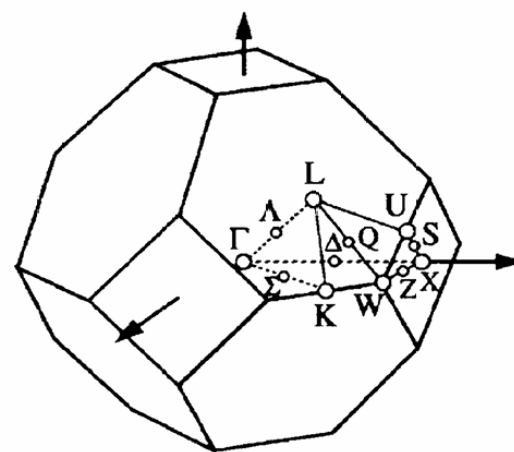
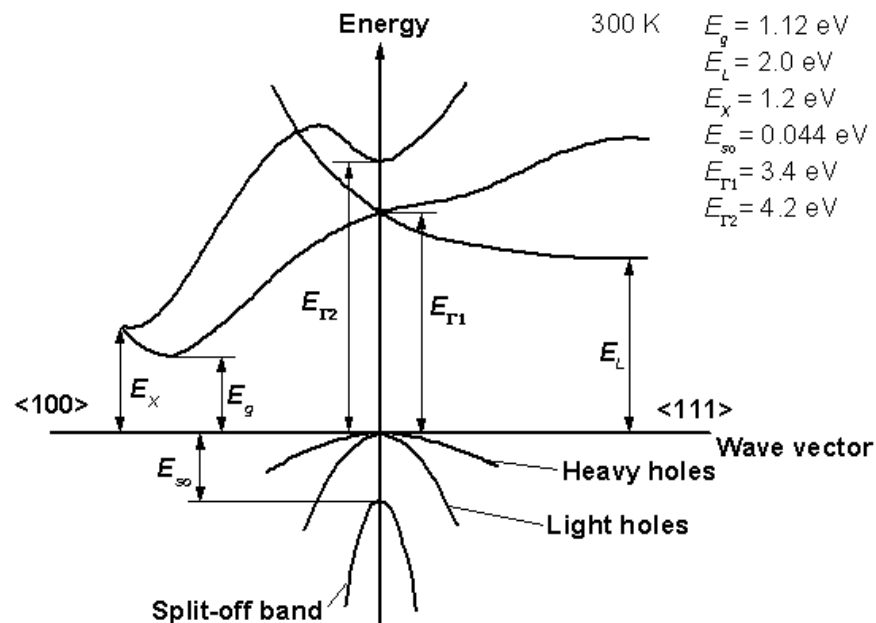
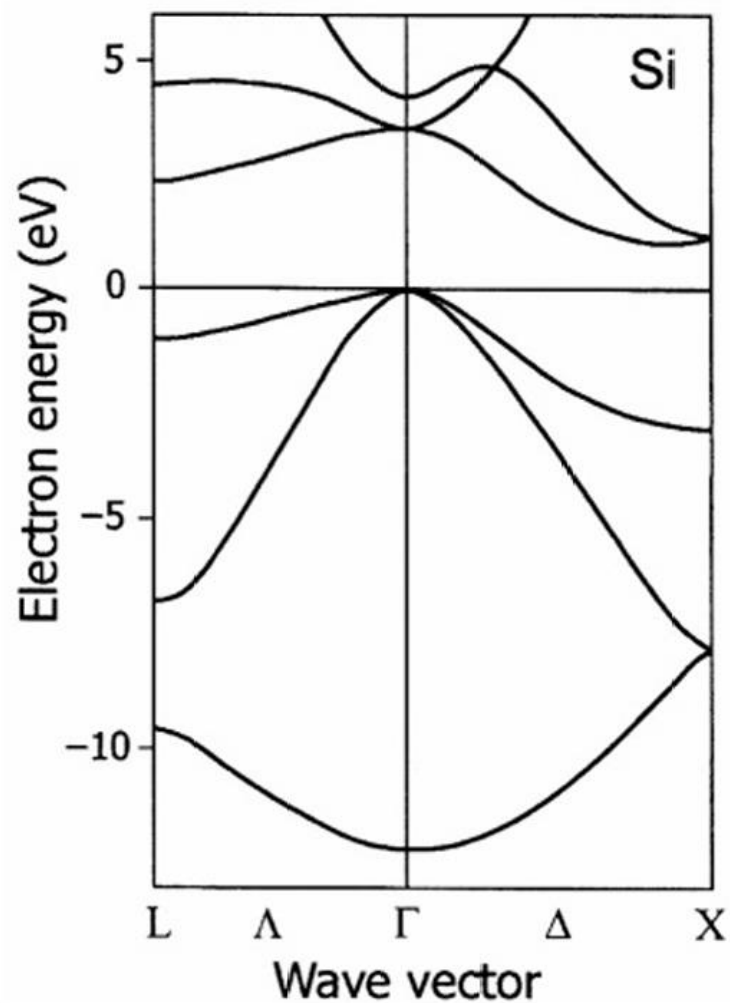
Electron-Lattice Interaction

- Frohlich interaction: the interaction of carriers with the electric field of preferentially longitudinal optical phonons.
- Deformation-potential scattering: the resulting interaction between carriers and phonons $\longleftrightarrow$ influence the band structure. So:

$$\Delta E = \Xi \frac{\Delta a}{a} \quad (8.21)$$

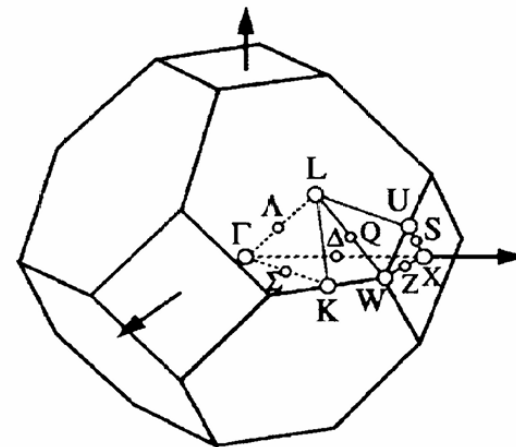
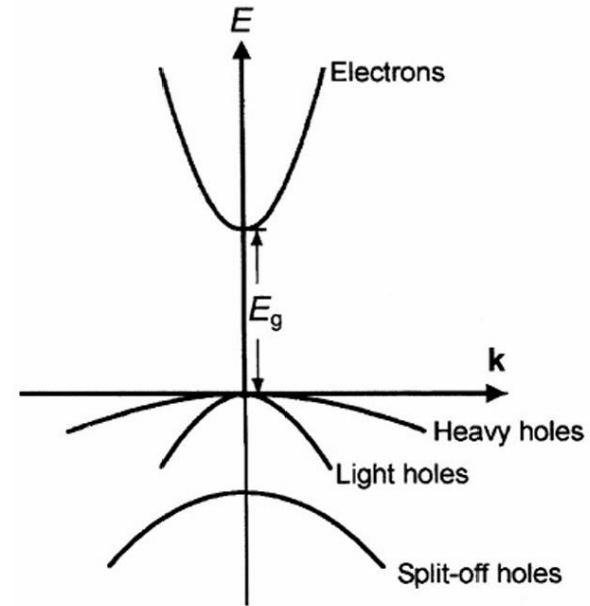
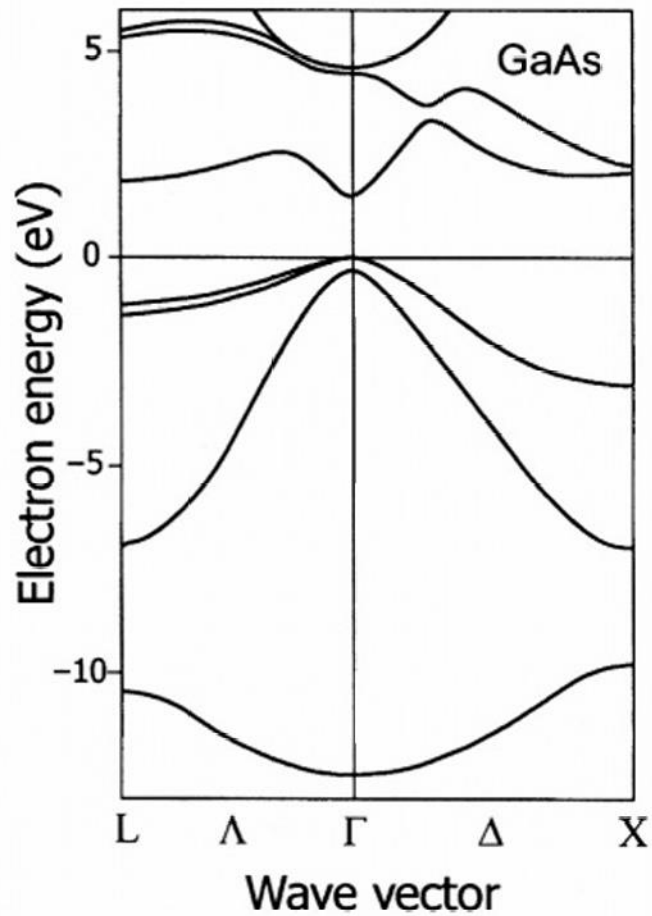
- Piezo (acoustic) coupling – Ultrasonic wave.

8.8 Bandstructures of Real Semiconductors



Bandstructure of Silicon

Band structure of GaAs



8.9 Density of States, Occupation Probability and Critical Points

- Density of states $D(E)$ for crystal electrons and holes using simple parabolic bands in the effective mass approximation, i.e.,

$$\text{conduction band (CB): } E(\mathbf{k}_e) = E_g + \frac{\hbar^2 \mathbf{k}_e^2}{2m_e} \quad (8.37a)$$

$$\text{valence band (VB): } E(\mathbf{k}_h) = \frac{\hbar^2 \mathbf{k}_h^2}{2m_h} \quad (8.37b)$$

So the density of the states depends on the energy in the conduction and valence bands:

$$\text{CB: } D(E) = (E - E_g)^{d/2-1}; \quad E > E_g \quad (8.38a)$$

$$\text{VB: } D(E) = E^{d/2-1}; \quad E > 0; \quad d = \text{dimensionality} \quad (8.38b)$$

The Fermi-Dirac statistics for electrons and holes:

$$\text{electrons: } f_{FD}(E) = \left(e^{\frac{(E-E_F^e)}{(k_B T)}} + 1 \right)^{-1} \quad (8.39a)$$

$$\text{holes: } f_{FD}(E) = 1 - \left(e^{\frac{(E-E_F^h)}{(k_B T)}} + 1 \right)^{-1} = \left(e^{\frac{-(E-E_F^h)}{(k_B T)}} + 1 \right)^{-1} \quad (8.39b)$$

In thermodynamic equilibrium:

$$E_F^e = E_F^h = E_F \quad (8.40a)$$

For parabolic bands with a degeneracy g , $d=3$,
total density of electrons in the conduction band n :

$$n = \frac{N}{L^3} = \int_{E_g}^{\infty} g_e \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{\frac{3}{2}} (E - E_g)^{\frac{1}{2}} \left(e^{\frac{(E-E_F^e)}{(k_B T)}} + 1 \right)^{-1} dE$$

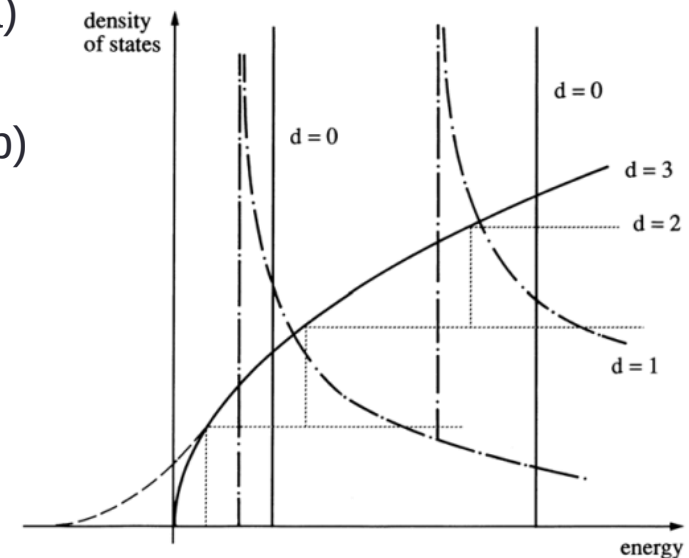


Fig. 8.20. Schematic drawing of the density of states as a function of energy for three-, and quasi-two-, one- and zero-dimensional systems in the effective mass approximation. The dashed line corresponds to disorder-induced localized states (Sect. 8.15)

- When E_F lies in the gap, by the Boltzmann function:

$$n = g_e \left(\frac{m_e k_B T}{2\pi \hbar^2} \right)^{\frac{3}{2}} e^{-\frac{(E_g - E_F)}{(k_B T)}} = n_{eff}(T, m_e) e^{-\frac{(E_g - E_F)}{(k_B T)}} \quad (8.42a)$$

If the electron gas is degenerate, i.e., if E_F is situated in the conduction band:

$$n = \frac{2}{6\pi^2} g_e \left(\frac{2m_e}{\hbar^2} \right)^{3/2} (E_F - E_g)^{3/2} \quad \text{for } T = 0 \text{ K.} \quad (8.42c)$$

- In analogy we find for a three-dimensional non-degenerate gas of holes:

$$p = p_{eff} e^{-E_F/k_B T} \quad p_{eff} = g_h \left(\frac{m_h k_B T}{2\pi \hbar^2} \right)^{3/2}. \quad (8.42d)$$

- In thermodynamic equilibrium :

$$np = n_{eff} p_{eff} e^{\frac{-E_g}{k_B T}} =: n_i^2(T) \quad (8.43)$$

Out of thermal equilibrium, one has:

$$n \cdot p \neq n_i^2(T) \quad \text{and} \quad \mu_{eh} = E_F^e - E_F^h \neq 0 \quad (8.44)$$

The so-called critical points or van Hove singularities:

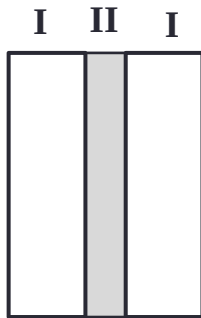
$$D(E): \begin{cases} (E - E_0)^{\frac{d}{2}-1} & \text{for } E \geq E_0 \\ 0 & \text{for } E < E_0 \end{cases} \quad M_0 \text{ critical point} \quad (8.45a)$$

If a parabolic band ends at a certain energy E_0 like a valence band, we have the inverse situation, i.e.,

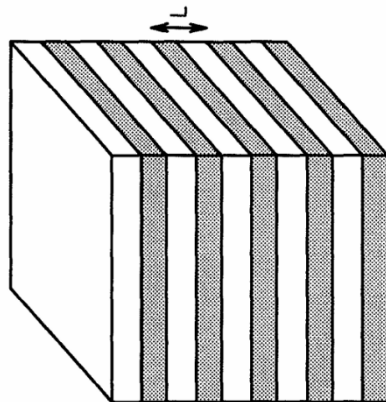
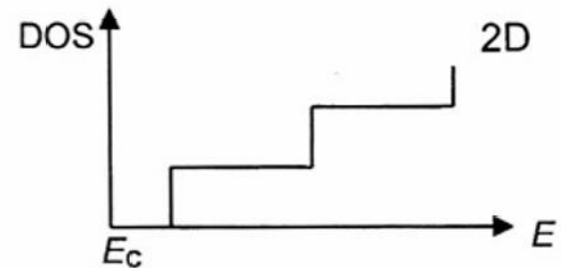
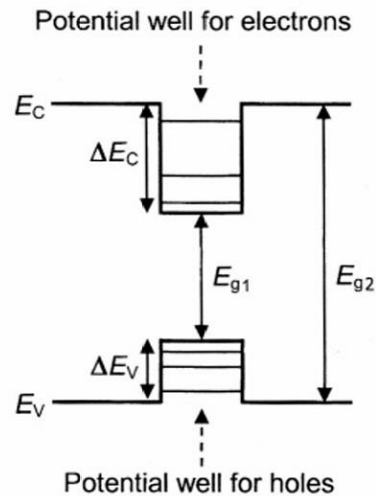
$$D(E): \begin{cases} (E_0 - E)^{\frac{d}{2}-1} & \text{for } E \leq E_0 \\ 0 & \text{for } E > E_0 \end{cases} \quad M_d \text{ critical point} \quad (8.45b)$$

8.10 Electrons and Holes in Quantum Wells and Superlattices

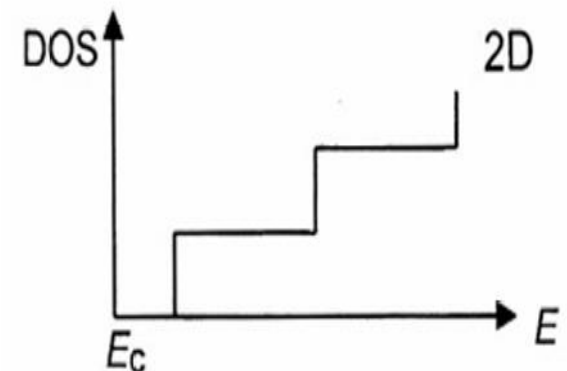
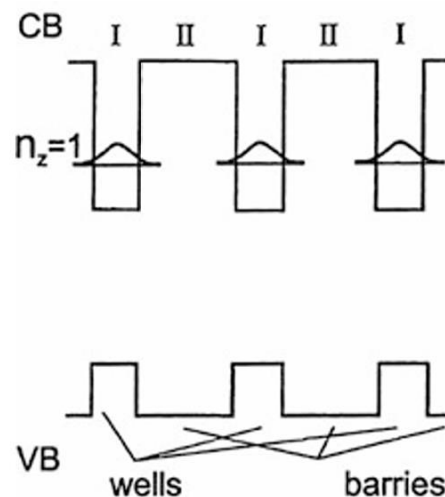
- Single Quantum well (SQW) and Multiple Quantum Wells (MQWs)



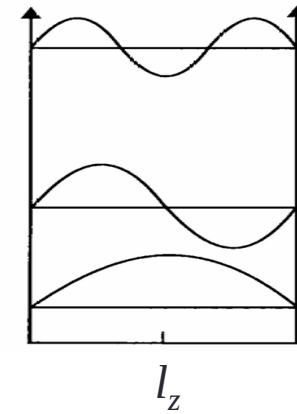
SQW



MQWs



Quantum wells with finite or infinite barrier height



QW with infinite barrier height

$$\left[\frac{\hat{p}^2}{2m} + V(\mathbf{r})\right]\phi(\mathbf{r}) = E\phi(\mathbf{r}) \quad \hat{p} = -i\hbar\nabla$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial z^2} \phi_{n_z}(z) = E_{n_z} \phi_{n_z}(z)$$

The solution of this problem with cos or even n_z :

$$\phi_{n_z}(z) = \cos(k_z z) \quad \text{for even parity} \quad \phi_{n_z}(z) = \sin(k_z z) \quad \text{for odd parity} \quad k_z = \sqrt{2mE_{n_z}/\hbar^2}$$

$$\psi_n(\mathbf{r}) = \phi_{n_z}(z) e^{i\mathbf{k}_{\parallel}\mathbf{r}_{\parallel}} = V^{-\frac{1}{2}} e^{i(k_x x + k_y y)} \cos(k_z z) \quad (8.47a)$$

$$\psi_n(\mathbf{r}) = \phi_{n_z}(z) e^{i\mathbf{k}_{\parallel}\mathbf{r}_{\parallel}} = V^{-\frac{1}{2}} e^{i(k_x x + k_y y)} \sin(k_z z) \quad (8.47b)$$

The energy of the particle:

$$k_z \cdot 2l_z = n_z 2\pi \quad k_z = \frac{n_z \pi}{l_z} \quad n_z = 1, 2, 3, \dots$$

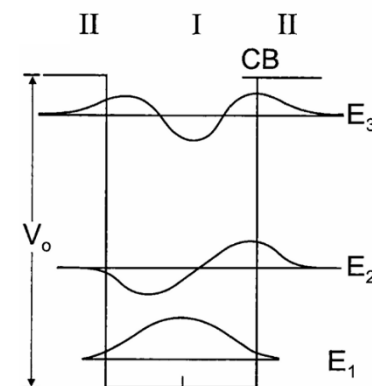
$$E_n(\mathbf{k}_{\parallel}) = \frac{\hbar^2 \mathbf{k}_{\parallel}^2}{2m_e} + E_{n_z} = \frac{\hbar^2 \mathbf{k}_{\parallel}^2}{2m_e} + \frac{\hbar^2 \pi^2 n_z^2}{2m_e l_z^2} = \frac{\hbar^2}{2m_e} \left(\mathbf{k}_{\parallel}^2 + n_z^2 \frac{\pi^2}{l_z^2} \right) \quad (8.47c)$$

For the more realistic case of a QW with finite potential height V_0 , the wave functions penetrate exponentially decaying into the barrier.

The Schrodinger equation we obtain for the wave function in z direction:

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial z^2} \phi_{n_z}(z) = E_{n_z} \phi_{n_z}(z) \quad |z| \leq \frac{l_z}{2}$$

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial z^2} + V(z) \right] \phi_{n_z}(z) = E_{n_z} \phi_{n_z}(z) \quad |z| \geq \frac{l_z}{2}$$



$$\phi_{n_z}(z) = \begin{cases} A \cos(k_z z) & \text{for even parity} \\ A \sin(k_z z) & \text{for odd parity} \end{cases} \quad |z| \leq \frac{l_z}{2} \quad k_z = \sqrt{2m^I E_{n_z} / \hbar^2} = \frac{n_z \pi}{l_z}$$

$$n_z = 1, 2, 3 \dots$$

$$\phi_{n_z}(z) = \begin{cases} B \exp(k'_z z) & z \leq -\frac{l_z}{2} \\ B \exp(-k'_z z) & z \geq \frac{l_z}{2} \end{cases} \quad k'_z = \sqrt{2m^{II}(V_0 - E) / \hbar^2}$$

Quantized energy levels for heavy hole

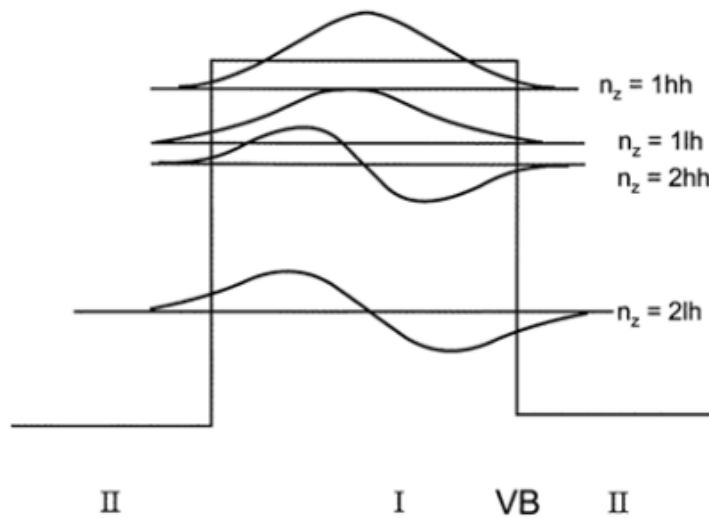


Fig. 8.23. The first two quantized levels for heavy and light holes (schematic)

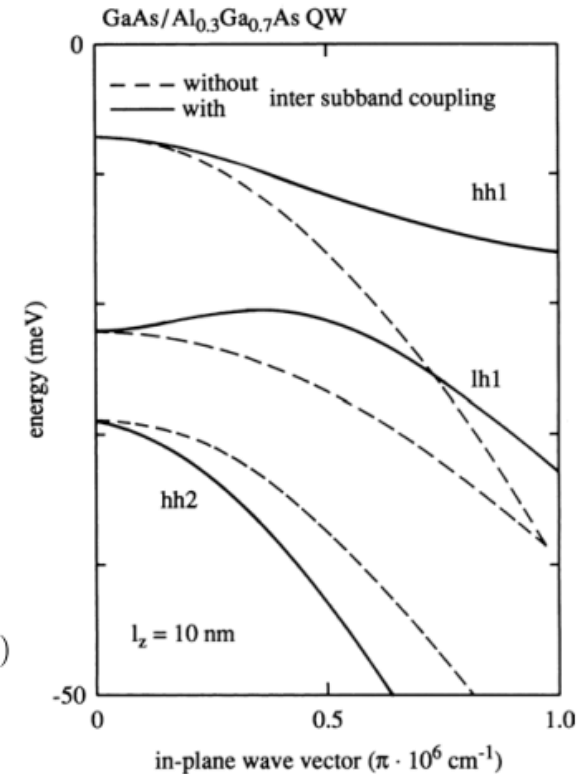


Fig. 8.24. The in-plane dispersion relation of the first three valence subbands in a GaAs/Al_{1-y}Ga_yAs quantum well without (---) and with (—) intersubband interaction ([86B1] or [01L1] of Chap. 1)

- Now introduce the terms multiple quantum well (MQW), coupled quantum wells and superlattice (SL) and discuss the various types of band alignment.

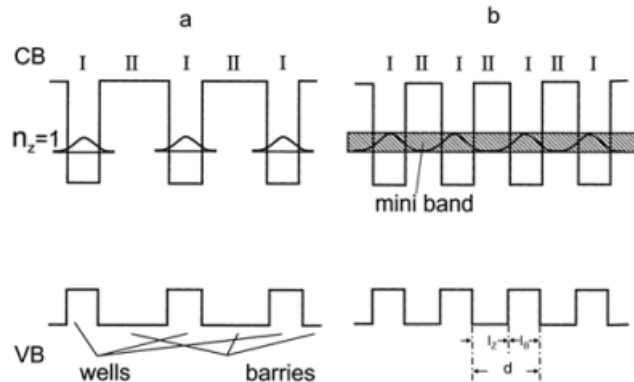


Fig. 8.25. The spatial variation of CB and VB extrema of a multiple quantum well (MQW) (a) and of a superlattice (SL) (b)

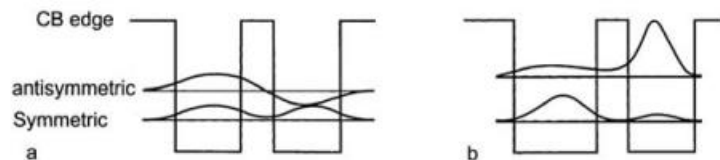


Fig. 8.26. Two coupled identical (a) and different (b) quantum wells and the envelope functions of the lowest states (schematic)

- If the hole mass is heavier, the penetration of the wave function into the barrier and the width of the bands are smaller.

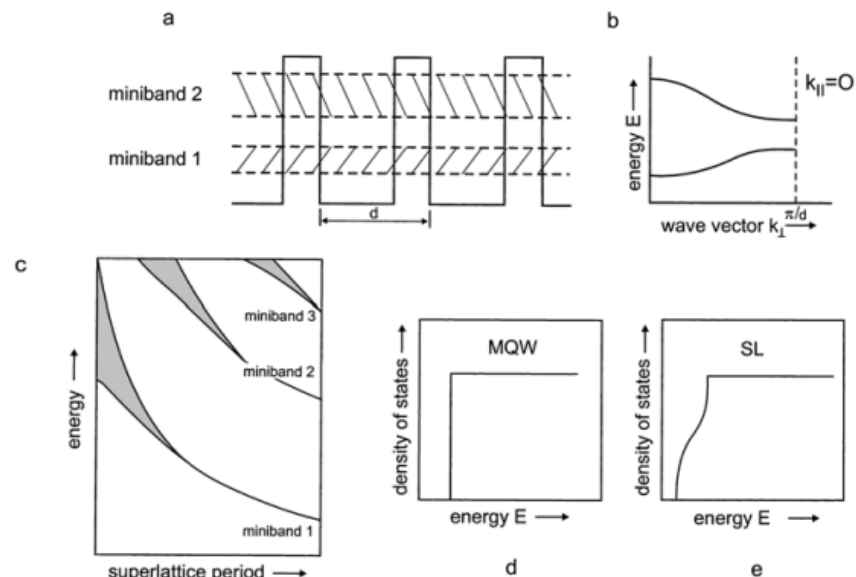


Fig. 8.27. Mini bands in the z direction ($k_x = k_y = 0$) of a SL (a, b) the width and position of these mini bands as a function of the superlattice period (c) and the density of states (e) compared to the one of MQW (d). According to [99B1,01L1] of Chap. 1

- Three types structures shown in Fig.8.28:

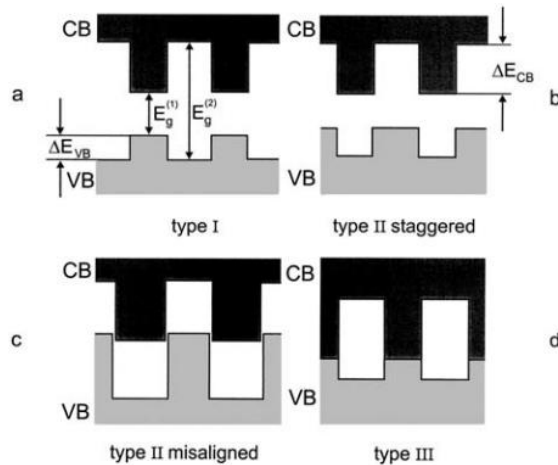


Fig. 8.28. Various band alignments in MQW or SL: type I (a), type II staggered (b), type II misaligned (c), type III (d)

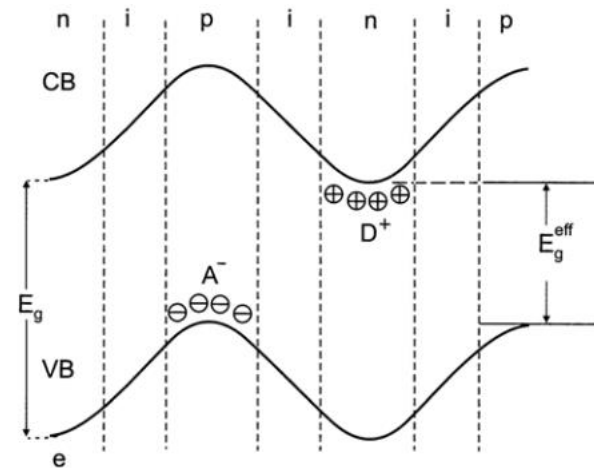


Fig. 8.29. The band alignment in a doping superlattice or nipi structure

- nipi structures or doping superlattices : with an almost parabolic confinement potential.

By solving the Poisson equation for the electrostatic potential : $\nabla^2 \varphi(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\epsilon \epsilon_0}$

Such structures are very sensitive, but slow photodetectors.

8.11 Growth of Quantum Wells and of Superlattices

- The two most important ways: MBE and MOCVD.

■ Molecular Beam Epitaxy (MBE)

- Relies on the sublimation of ultrapure elements, then condensation of them on wafer
- In a vacuum chamber (pressure: $\sim 10^{-11}$ Torr).
- “Beam”: molecules do not collide to either chamber walls or existent gas atoms.
- Growth rate: $1\mu\text{m/hr}$.

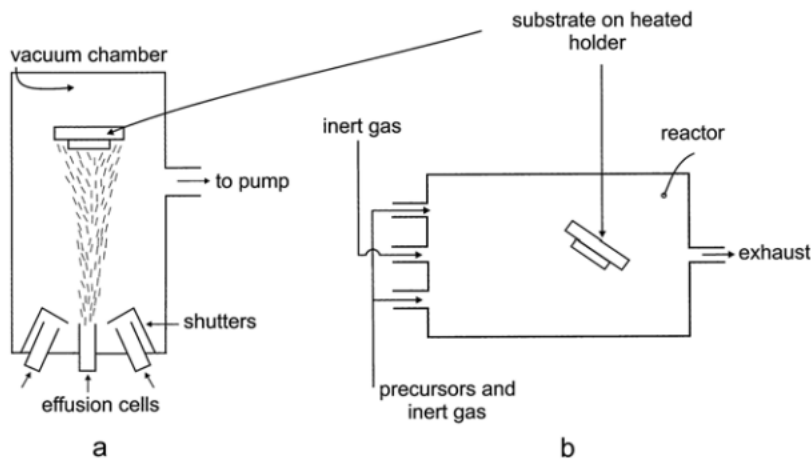
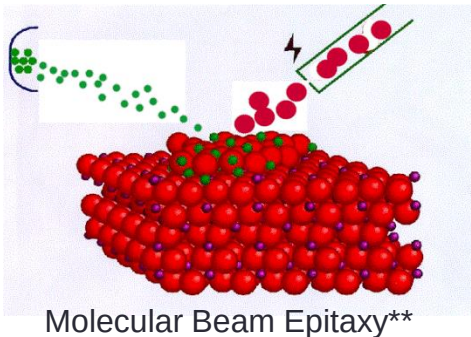
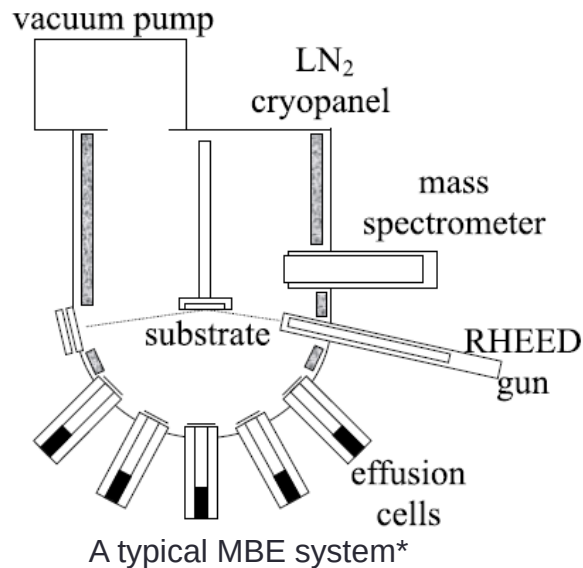


Fig. 8.30. The schematics of a MBE chamber (a) and of a MOCVD reactor (b)



MBE: Working Principle



- Epitaxial growth: Due to the interaction of molecular or atomic beams on a surface of a heated crystalline substrate.
 - The solid source materials sublime
 - They provide an angular distribution of atoms or molecules in a beam.
 - The substrate is heated to the necessary temperature.
 - The gaseous elements then condense on the wafer where they may react with each other.
- Atoms on a clean surface are free to move until finding correct position in the crystal lattice to bond.
- Growth occurs at the step edges formed: More binding forces at an edge.

MOCVD: Working Principle

- Growth of single crystals of GaAs by Manasevit, in 1968
- Source: trimethylgallium(TMGa)–arsine(AsH_3)– H_2
- Pyrolyzed temp: 600~700 °C
- Substrate : GaAs ,sapphire or others
- Basic reaction: $\text{Ga}(\text{CH}_3)_3 + \text{AsH}_3 \rightarrow \text{GaAs} + 3\text{CH}_4$

? Which materials can be grown epitaxially on top of each other with good quality

- Equal or similar crystal structures
- Almost identical lattice constants
- The same valences
- Similar types of chemical binding

Homoepitaxy and heteroepitaxy

- Homoepitaxy : grow on a substrate the same material
- Heteroepitaxy : substrate and growing layers are different

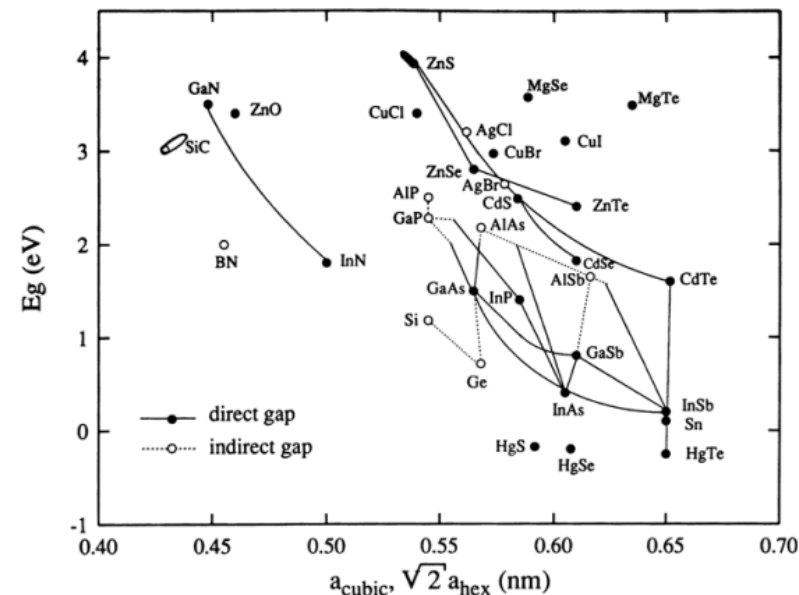


Fig. 8.31. Width of the band gaps as a function of the lattice constant for various semiconductors and their alloys. *Solid lines:* direct gap; *dashed lines:* indirect gap. Note that this Figure gives information about the band gap only, but not about the relative alignment, i.e., whether the combination will be of type I, II or III. Data from [82L1,01L1] of Chap. 1

- The usual behavior under tensile and compressive strain is shown in Fig. 8.32.

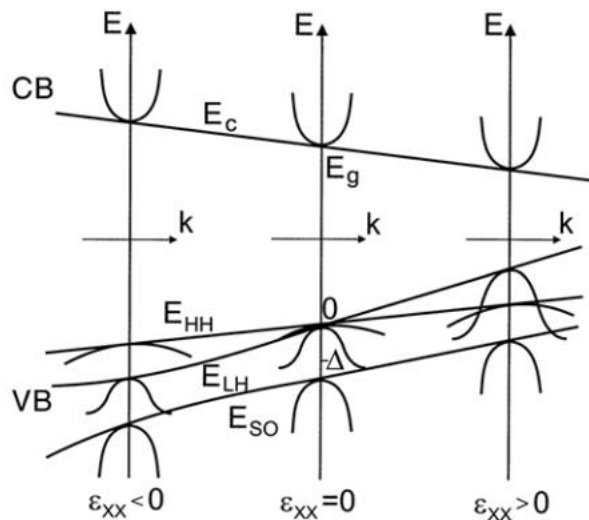


Fig. 8.32. Schematic drawing of the lhh - lh splitting, of the spin orbit splitting and of the change of the band gap under two-dimensional compressive ($\epsilon_{xx} < 0$) or tensile strain ($\epsilon_{xx} > 0$) [93T1]

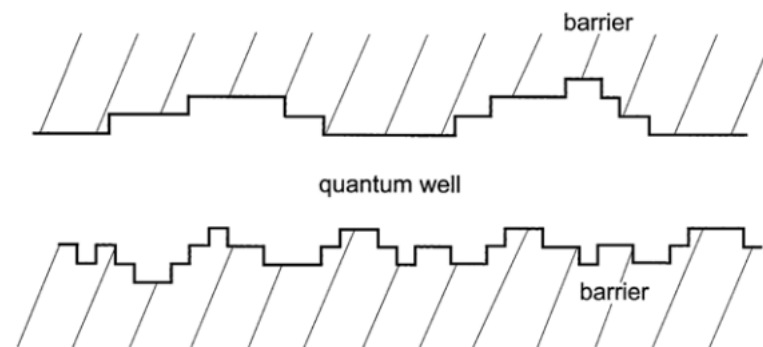


Fig. 8.33. Thickness fluctuations of a quantum well with two different length scales on both sides (schematic)

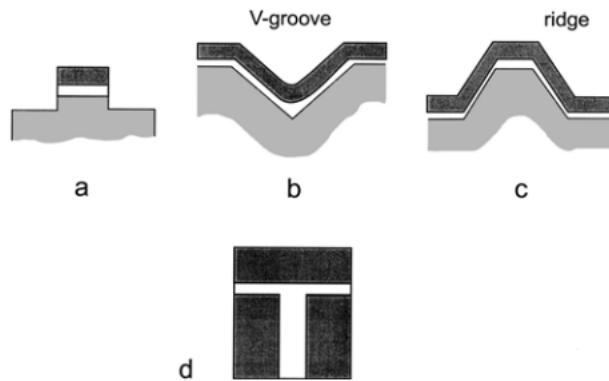
Some complications encountered in real quantum wells and superlattices in Fig.8.33

8.12 Quantum Wires

- Quasi-one-dimensional or quantum wires : only free motion in one direction.
- The energies of the carriers :

$$E(k_x) = E_{ni,nj} + \frac{\hbar^2 k_x^2}{2m_{e,h}} \quad (8.52)$$

- The quantization energy $E_{ni,nj}$ depends on the cross section ,of the wire.



1. by lithographical and etching of quantum wells
2. create pre-patterned substrates with either V grooves or ridges
3. cleaved-edge overgrowth

Fig. 8.34. Various possibilities to create quasi-one-dimensional quantum wires. Photolithography and etching of quantum wells (a) growth in V-grooves (b) on ridges (c) or T-shaped quantum wires occurring at the junction of two orthogonal quantum wells. In all cases the wire extends normal to the paper

- For this rectangular cross section the confinement in two directions is the one of the well E_{nz} .
- Typically: $3nm \leq l_z \leq 10nm$ and $20nm \leq l_y \leq 50nm$
- Usually : $E_{nz} \gg E_{ny}$ for $nz = ny$

8.13 Quantum Dots

- Quantum Dots: confinement in all three directions of space leaving no more possibility for free propagation of the carriers.
- The energy levels in a single quantum dot are discrete: $E = E_{nk,nj,nk}$ (8.55)

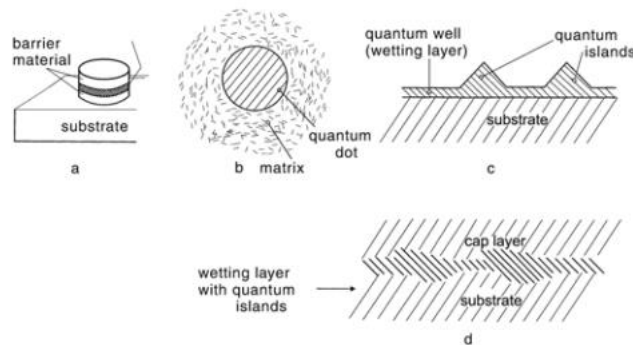


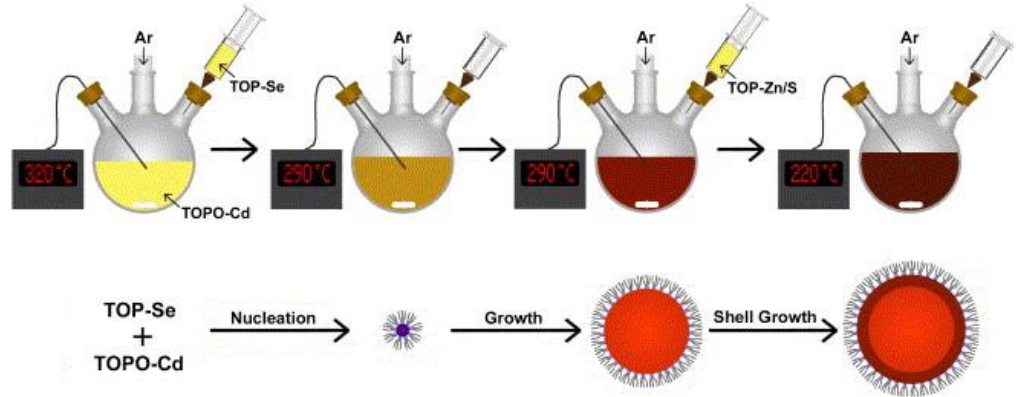
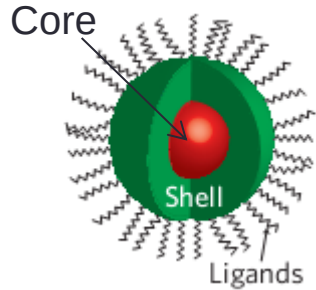
Fig. 8.35. Various possibilities to form quantum dots by microstructuring of quantum wells (a), by precipitation in glasses or organic solutions (b) and by “self-assembly” (c, d) (schematic)

1. by lithographical and etching of micro- or nanostructuring quantum wells
2. by an annealing procedure in glasses doped with the constituents of semiconductor
3. chemical precipitation
4. “self-assembled” quantum dots

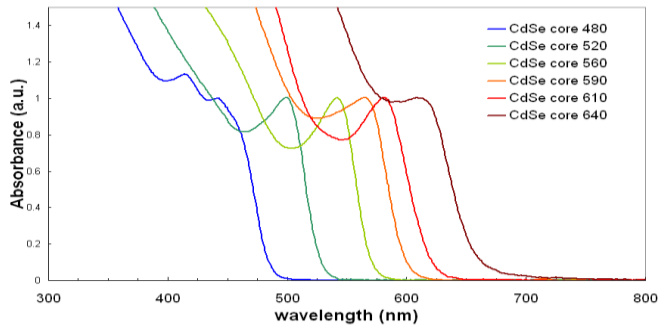
- Small changes to the size or composition of a quantum dot allow the energy levels, and the band-gap, to be fine-tuned to specific, desired energies.

Colloidal quantum dots

- Colloidal Quantum Dots (QDs) are solution processable nanoscale crystals of semiconducting materials



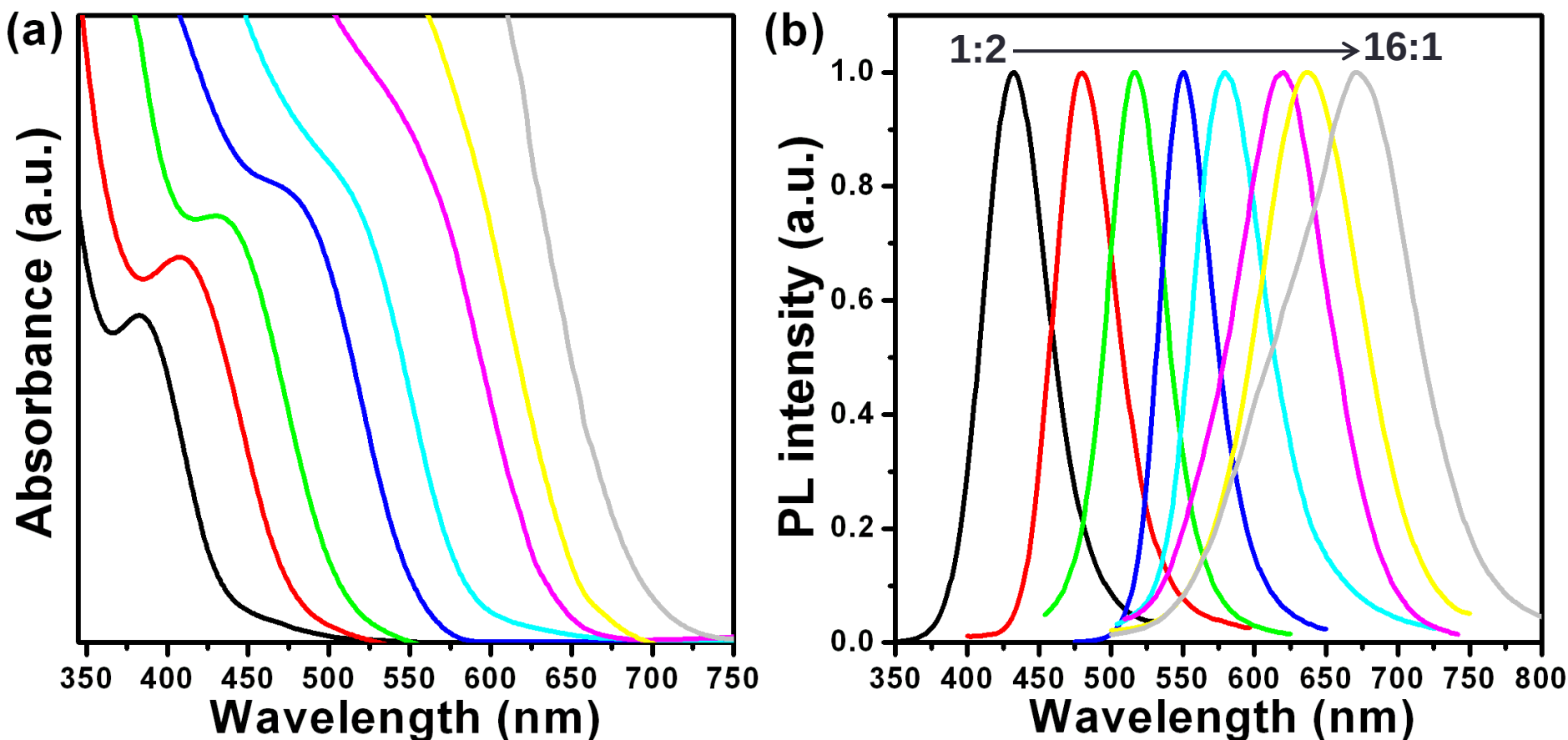
- Size dependent optical properties



- Narrow emission (FWHM ~ 30 nm)

- High efficiency (PL efficiency ~ 100%)
- Solution processable
- Stable than organic semiconductors
- Broad applications
 - Light-emitting diodes (LEDs)
 - Solar cells
 - Photodiodes
 - Field-effect transistors (FETs)
 - Bio-imaging, sensing and labelling

Abs & PL spectra



UV-Vis absorption and PL spectra of the resulting InP/ZnS NCs with different ratios of InP:ZnS (increasing the ratio of InP to ZnS from left to right). The PL and absorption spectra of NCs were red-shifted successively.

8.14 Defects, Defect States and Doping

Imperfections :

- Point defects
- One dimensional defects like dislocations
- 2D: Stacking faults, grain boundaries, surfaces or interfaces
- 3D: Precipitates or voids

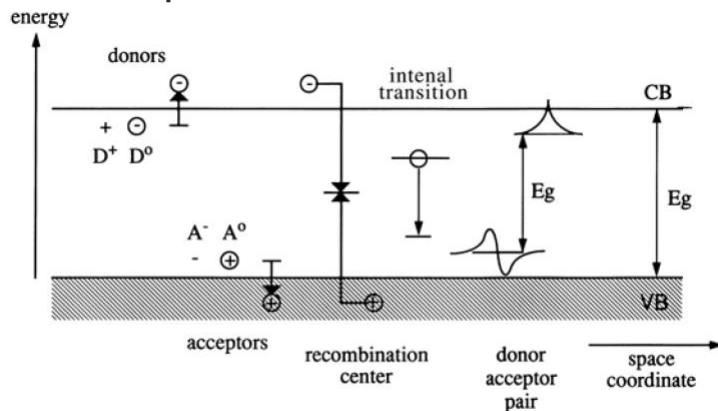


Fig. 8.36. Schematic drawing of various impurity levels in semiconductors

Donors: $D^0 \rightleftharpoons D^+ + e^-$, like N or P in Ge or Si

Acceptors: $A^0 \rightleftharpoons A^- + h^+$, like Ga or B in Si and Ge

The same atom can act as donor or acceptor depending on the way it is introduced into the lattice.

A shallow donor (acceptor) can be considered as a positively (negatively) charged center to which an electron (hole) is bound by Coulomb interaction.

- Binding energy: $E_b^{D,A} = Ry \frac{m_{e,h}}{m_0} \frac{1}{\varepsilon^2} \frac{1}{n_B^2}$ (8.57a)

- Depending on the material parameters , binding energies of donor and acceptor:

$$5meV \leq E_b^D \leq 50meV \quad (8.57b)$$

$$20meV \leq E_b^A \leq 200meV \quad (8.57c)$$

- The wave function of a shallow donor (or acceptor) can be described as :

$$\phi(\mathbf{r}) = \sum_k a_k \phi_k^{CB}(\mathbf{r}) \quad (8.58)$$

- The occupation probability of donors and/or acceptors in thermodynamic equilibrium:

$$f_{D/A} = \frac{1}{1 + \frac{1}{g_t \exp\left(\frac{(E_D + E_F)}{k_B T}\right)}} \quad (8.59)$$

- Two special methods of doping connected with (M)QW:

- δ -doping
- Modulation doping

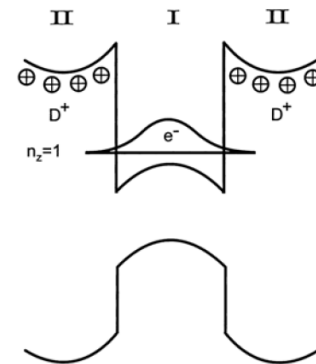


Fig. 8.37. The curvature of the band edges in the case of symmetric modulation doping of a MQW (schematic) including the first quantized electron level

- Deep donors and acceptors and a variety of other deep centers: have one or more energy levels somewhere around the middle of the gap .

8.15 Disordered Systems and Localization



- What new in disordered systems?
- How disorder can be realized?

Disordered models: Anderson model and Lifshitz model

$BV_0^{-1} > 1$, Both localized states and extended states

$BV_0^{-1} < 1$, Only localized states

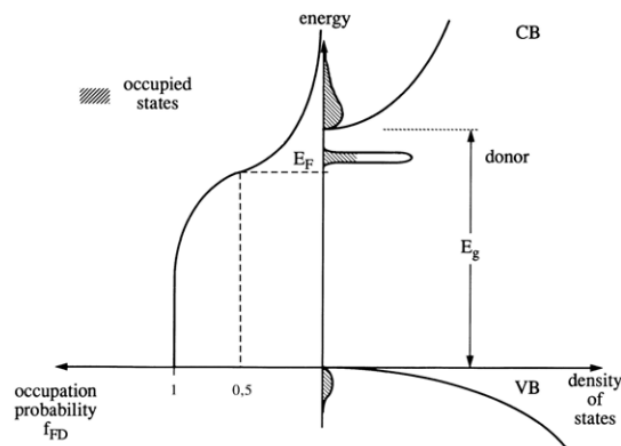


Fig. 8.38. The density of states DOS in the effective-mass approximation of an idealized three-dimensional semiconductor containing some impurities, here a donor state, and the resulting population (hatched areas). Usually the relation $k_B T \ll E_g$ holds and the Fermi-function looks more step-like with the consequence that the populations in conduction and valence bands cannot be shown simultaneously on the same linear scale

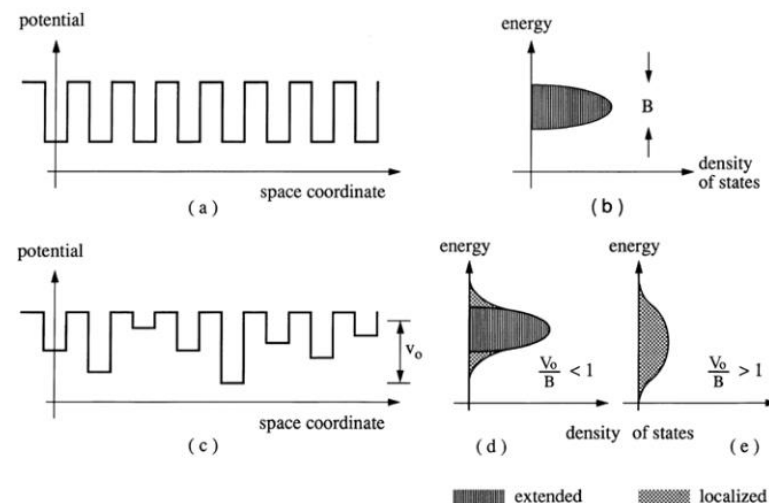


Fig. 8.39. A periodic potential (a) and the resulting band, represented by its density of states (b), a disordered system (c) and bands with localized and delocalized states (d) and with localized states only (e)

- ❑ 1958年，美国物理学家安得森首先预测，如果在导体内加入杂质，电子在传导时会被这些杂质散射，多重散射波则发生互相干扰，结果能导致电子的运动停止，金属的导电性消失，呈现出绝缘体的性质。
- ❑ 这一预测后来被实验证实，安得森也因此荣获诺贝尔物理奖。现在人们称此由于掺杂而导致的导电到绝缘的现象为电子波的安得森局域化，即电子局域化。



- The explain of extent wave and localized wave:

$$\phi_{\mathbf{k}}^{Bloch}(\mathbf{r}) = \sum_i e^{i\mathbf{k}\cdot\mathbf{R}_i(\mathbf{r}-\mathbf{R}_i)} = e^{i\mathbf{k}\cdot\mathbf{r}} \sum_i e^{-i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R}_i)} \phi_n(\mathbf{r}-\mathbf{R}_i) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r})$$

In an extended state :

$$\phi^{ext}(\mathbf{r}) = \sum_i c_i \phi_n(\mathbf{r}-\mathbf{R}_i)$$

For a localized state:

$$\phi^{loc}(\mathbf{r}) = \sum_i d_i \phi_n(\mathbf{r}-\mathbf{R}_i) e^{-\frac{|\mathbf{r}-\mathbf{R}_i|}{\xi}}$$

- A disordered system is in principle characterized only by its density of states $D(E)$.
- The realizations of disorder in semiconductors:
 - Heavily doped semiconductors
 - Alloy semiconductors
 - Well-width fluctuations
 - Amorphous semiconductors

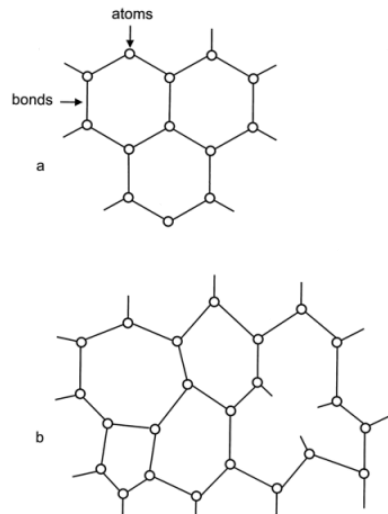


Fig. 8.42. A hypothetical two-dimensional crystalline (a) and amorphous (b) material

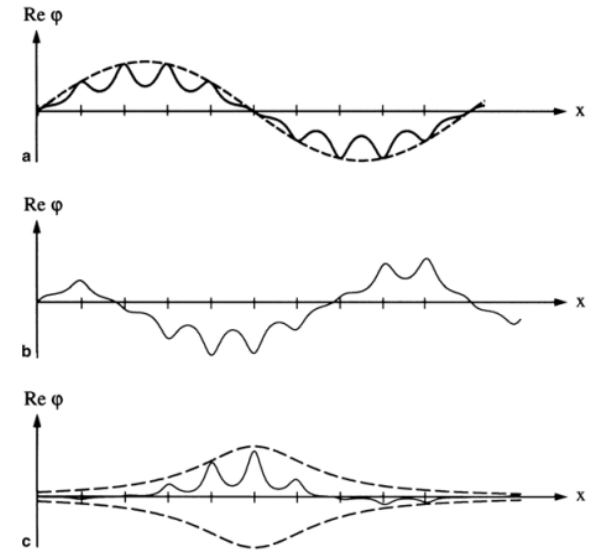


Fig. 8.40. Schematic drawing of the real part of the wave function for a periodic potential (a) and for a disordered system an extended state (b) and a localized one (c)

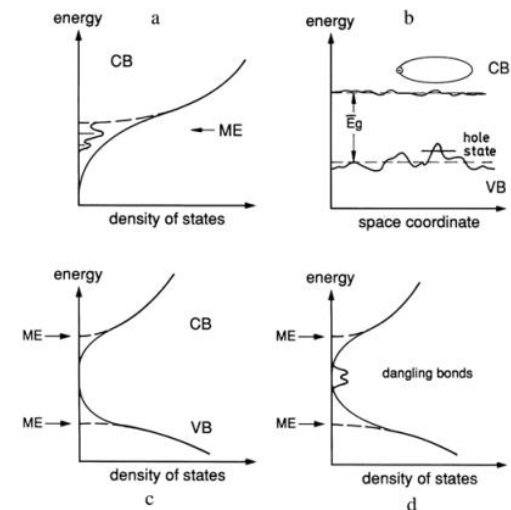


Fig. 8.41. The appearance of tails of localized states due to increasing doping (a), potential fluctuations, e.g., in an alloy (b, c) or in amorphous semiconductors (d)